How-To Library

Informatica

Frequently Asked Questions for Google BigQuery Connector

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Abstract

This article describes frequently asked questions about using Google BigQuery V2 Connector to read data from and write data to Google BigQuery.

Supported Versions

- Cloud Data Integration

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General Questions

What is Google Cloud Platform?

Google Cloud Platform is a set of public cloud computing services offered by Google. It provides a range of hosted services for compute, storage, and application development that run on Google hardware. Google Cloud Platform services can be accessed by software developers, cloud administrators, and other enterprise IT professionals over the public internet or through a dedicated network connection.

Google Cloud Platform provides Google BigQuery to perform data analytics on large datasets.

How can I access Google Cloud Platform?

You must create a Google service account to access Google Cloud Platform. To create a Google service account, click the following URL: https://cloud.google.com/

What are the permissions required for the Google service account to read data from or write data into a table in a Google BigQuery dataset?

You must verify that you have read and write access to the Google BigQuery dataset that contains the source table and target table.

When you read data from or write data to a Google BigQuery table, you must have the following permissions:

- bigquery.datasets.create
- bigquery.datasets.get
- bigquery.datasets.getIamPolicy
- bigquery.datasets.updateTag
- bigquery.models.*
- bigquery.routines.*
- bigquery.tables.create
- bigquery.tables.delete
- bigquery.tables.export
- bigquery.tables.get
- bigquery.tables.getData
- bigquery.tables.list
- bigquery.tables.update

- bigquery.tables.updateData
- bigquery.tables.updateTag
- resourcemanager.projects.get
- resourcemanager.projects.list
- bigquery.jobs.create

When you only read data from a Google BigQuery table, you must have the following permissions:

- bigquery.datasets.get
- bigquery.datasets.getIamPolicy
- bigquery.models.getData
- bigquery.models.getMetadata
- bigquery.models.list
- bigquery.routines.get
- bigquery.routines.list
- bigquery.tables.export
- bigquery.tables.get
- bigquery.tables.getData
- bigquery.tables.list
- resourcemanager.projects.get
- resourcemanager.projects.list
- bigquery.jobs.create
- bigquery.tables.create

What are the operations that Data Integration supports for Google BigQuery V2 Connector?

You can use Google BigQuery V2 Connector to read from and write data to Google BigQuery. You can perform the following operations when you write data to a Google BigQuery target:

- Insert
- Update
- Upsert
- Delete

What are the operations that I can perform on partitioned tables using Google BigQuery V2 Connector?

You can perform the insert operation on partitioned tables using Google BigQuery V2 Connector. For more information about partition tables, click the following URL:
https://cloud.google.com/bigquery/docs/reference/standard-sql/data-manipulation-language

Does Google BigQuery V2 Connector support both the Data Integration Hosted Agent and the Secure Agent?

Yes. Google BigQuery V2 Connector supports both the Data Integration Hosted Agent and the Secure Agent.

What information do I need to create a Google BigQuery connection?

Before you create a Google BigQuery connection, you must have the following information:

- Runtime Environment. Name of the runtime environment where you want to run the tasks.
- Service Account ID. The client_email value present in the JSON file that you download after you create a Google service account.

- Service Account Key. The private_key value present in the JSON file that you download after you create a Google service account.
- Project ID. The project_id value present in the JSON file that you download after you create a Google service account.
- Dataset ID. Name of the dataset that contains the source table and target table that you want to connect to.
- Storage Path. Path in Google Cloud Storage where the Google BigQuery V2 Connector creates a local stage file to store data temporarily.
- Connection mode. The mode that you want to use to read data from or write data to Google BigQuery. You can select Simple, Hybrid, or Complex connection modes.
- Schema Definition File Path. Specifies a directory on the client machine where the Data Integration Service must create a JSON file with the sample schema of the Google BigQuery table. The JSON file name is the same as the Google BigQuery table name. Alternatively, you can specify a storage path in Google Cloud Storage where the Data Integration Service must create a JSON file with the sample schema of the Google BigQuery table. You can download the JSON file from the specified storage path in Google Cloud Storage to a local machine.

What modes can I use to read data from a Google BigQuery source using Google BigQuery V2 Connector?

You can use staging or direct mode to read data from a Google BigQuery source using Google BigQuery V2 Connector.

What modes can I use to write data to a Google BigQuery target using Google BigQuery V2 Connector?

You can use bulk or streaming mode to write data to a Google BigQuery target using Google BigQuery V2 Connector.

While writing data, does Google BigQuery V2 Connector persist the staging file in Google Cloud Storage?

By default, the Google BigQuery V2 Connector deletes the staging file in Google Cloud Storage. However, you can configure the task or mapping to persist the staging file after writing data to the Google BigQuery target. To persist the staging file, select the **Persist Staging File After Loading** option in the advanced target properties.

What are the staging file data formats that Google BigQuery V2 Connector supports?

Google BigQuery V2 Connector supports CSV and JSON (Newline Delimited) data formats for the staging files created in Google Cloud Storage.

You can use CSV data format to for staging files that contain primitive data and non-repeated columns. You cannot use CSV data format for staging files that contain Record data and repeated columns.

What are the data types that Data Integration supports for Google BigQuery sources and targets?

Data Integration supports the following data types for Google BigQuery sources and targets:

- BOOLEAN
- BYTE
- DATE
- DATETIME
- FLOAT
- INTEGER
- RECORD
- STRING
- TIME

- TIMESTAMP

Does Google BigQuery V2 Connector support partitioning?

Yes. Google BigQuery V2 Connector supports key range partitioning for Google BigQuery sources.

You can configure a partition key for fields of the following data types:

- String
- Integer
- Timestamp. Use the following format: YYYY-MM-DD HH24:MI:SS

Note: You cannot configure a partition key for Record data type columns and repeated columns.

Can I use PowerExchange for Cloud Applications to read data from or write data to Google BigQuery?

Yes. You can use PowerExchange for Cloud Applications to read data from or write data to Google BigQuery.

Does Google BigQuery V2 Connector support federated queries?

Yes. You can use Google BigQuery V2 Connector or PowerExchange for Cloud Applications to read data from federated data sources such as Google Cloud Storage and Google Cloud Bigtable. Ensure that the data location of the Google BigQuery dataset, Google Cloud Bigtable, and Google Cloud Storage bucket is the same.

How do I access the sample schema of a Google BigQuery table?

Google BigQuery V2 Connector creates a JSON file with the sample schema of the Google BigQuery table. In the **Schema Definition File Path** field of the Google BigQuery connection, you can specify a directory where the Google BigQuery V2 Connector must create the JSON file.

Alternatively, you can specify a storage path in Google Cloud Storage where the Google BigQuery V2 Connector must create a JSON file with the sample schema of the Google BigQuery table. You can download the JSON file from the specified storage path in Google Cloud Storage to a local machine.

Performance Tuning Questions

When should I use direct mode to read data from a Google BigQuery source?

Use direct mode when the volume of data that you want to read is small. In direct mode, Google BigQuery V2 Connector directly reads data from a Google BigQuery source.

When should I use staging mode to read data from a Google BigQuery source?

Use staging mode when you want to read large volumes of data in a cost-efficient manner.

In staging mode, Google BigQuery V2 Connector first exports the data from the Google BigQuery source into Google Cloud Storage. After the export is complete, Google BigQuery V2 Connector downloads the data from Google Cloud Storage into a local stage file. You can configure the local stage file directory in the advanced source properties. Google BigQuery V2 Connector then reads the data from the local stage file.

When should I use bulk mode to write data to a BigQuery target?

Use bulk mode when you want to write large volumes of data and improve the performance. In bulk mode, Google BigQuery V2 Connector first writes the data to a staging file in Google Cloud Storage. When the staging file contains all the data, Google BigQuery V2 Connector loads the data from the staging file to the BigQuery target.

When should I use streaming mode to write data to a BigQuery target?

Use streaming mode when you want the Google BigQuery target data to be immediately available for querying and real-time analysis. In streaming mode, Google BigQuery V2 Connector directly writes data to the BigQuery target. Google BigQuery V2 Connector appends the data into the BigQuery target.

Evaluate Google's streaming quota policies and billing policies before you use streaming mode.

Which data format should I use for the staging file to improve the performance?

Use the CSV data format to improve the performance.

Should I enable staging file compression when I read data from a Google BigQuery source?

You can enable staging file compression to improve the performance.

Should I enable staging file compression when I write data to a Google BigQuery target?

You can enable staging file compression when the network bandwidth is low. Enabling compression reduces the time that Google BigQuery V2 Connector takes to write data to Google Cloud Storage. However, there will be a performance degradation when Google BigQuery V2 Connector writes data from Google Cloud Storage to the Google BigQuery target.

Does the Number of Threads for Downloading Staging Files option impact the performance when I read data from a BigQuery source in staging mode?

No. The value of the **Number of Threads for Downloading Staging Files** option does not affect the performance when you read data from a Google BigQuery source in staging mode. However, if you specify a value greater than one, the reader will consume more resources.

How do I configure the Number of Threads for Uploading Staging Files option to improve the performance?

To improve the performance when you write data to a Google BigQuery target in bulk mode, you can set the **Number of Threads for Uploading Staging File** option to 10. This recommendation is based on observations in an internal Informatica environment using data from real-world scenarios. The performance might vary based on individual environments and other parameters. You might need to test different settings for optimal performance.

If you configure more than one thread, you must increase the Java heap size in the **JVMOption3** field for DTM under the **System Configuration Details** section of the Secure Agent. You must then restart the Secure Agent for the changes to take effect.

How do I configure the local stage file directory when I write to a Google BigQuery target?

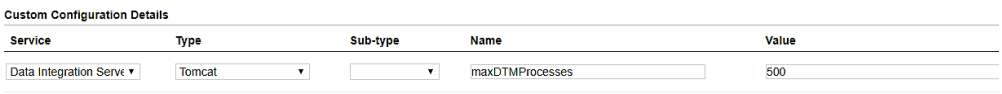
Set the local stage file directory to a directory on your local machine to improve the performance. By default, the local stage file directory is set to `/Temp` where the Secure Agent is installed.

How can I run concurrent mapping tasks?

To run concurrent mapping tasks, you must perform the following tasks:

1. Open **Administrator** and select **Runtime Environments**.
2. Select the Secure Agent that is used as the runtime environment for a mapping task.
3. Click **Edit Secure Agent** in **Actions**.
   The **Edit Secure Agent** page appears.
4. Select the **Service** as **Data Integration Server** in the **Custom Configuration Details** section.
5. Select the **Type** as **Tomcat** in the **Custom Configuration Details** section.
6. Add **maxDTMProcesses** in the **Name** field.
7. Specify a number in the **Value** field based on the number of concurrent mapping tasks that you want to run.
   For example, specify `500` in the **Value** field to run 500 concurrent mapping tasks.

   The following image shows the **Custom Configuration Details** section:



8. Repeat steps 2 through 7 on every Secure Agent that is used as the runtime environment for a mapping task.
9. Create a schedule on the **Schedules** page or in the mapping task wizard.
10. Associate the schedule with all the mapping tasks that you want to run concurrently.

Note: This recommendation is based on observations in an internal Informatica environment using data from real-world scenarios. The performance might vary based on individual environments and system configuration. You might need to test different settings for optimal performance. Google BigQuery limits the number of API calls made per service account. To avoid run-time failures, ensure that you have Google BigQuery connections with different service account IDs.

Author

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